

THE SHAPER

SHAPING MACHINE (THE SHAPER)

1.1 Introduction:

A shaper is a type of machine tool that uses linear relative motion between the work piece and a single-point cutting tool to machine a linear tool path. The shaper is a relatively simple machine. It is used fairly often in the tool room or for machining one or two pieces for prototype work. Tooling is simple, and shapers do not always require operator attention while cutting. A shaping machine is used to machine flat surface. It can cut grooves, angles and many other shapes. The Main Function of shaper is to generate a flat surface by combination of linear movement of cutting tool and work piece.

Shaping is where the work piece is fed at right angles to the cutting motion between successive strokes of the tool. Shaping and planners among the oldest techniques used and Shapers are where the work piece is fed at right angles to the cutting motion between successive strokes of the tool. In Planners the work piece is reciprocated and the tool is fed at right angles to the cutting motion. These processes require skilled operators and for the most part have been replaced by other processes

The horizontal shaper is the most common type and its principal components are shown below and described as follows:

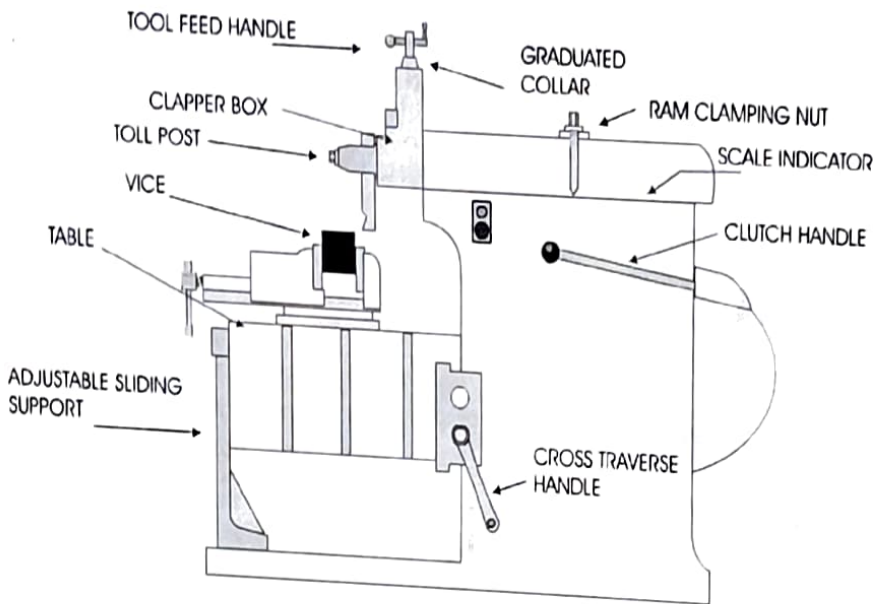


Fig.1 Horizontal Shaper Machine

1.2 Parts of Shaper:

Base:

It is a heavy and robust cast iron body. The base supports the column or pillar which supports all the working parts such as ram, work-table, drive mechanism etc.

Column or Pillar:

The column is a ribbed casting of cellular construction. The ram slide ways are provided on the top of the column while the table slide ways are machined on the front. The crank and slotted link mechanism that drives the ram is contained within the column. The

driving motor, variable speed gear box, levers, handles and other controls of shaper are also contained in the column.

Ram:

Ram is a rigidly braced casting and is located on the top of the column. The ram slides back and forth in dovetail or square ways to transmit power to the cutter. The starting point and the length of the stroke can be adjusted using stroke positioning mechanism and the down feed mechanism.

Tool head:

It is the device which holds the tool. The tool head slides in a dovetail at the front of the ram by means of T-bolt and is fastened to the ram on a circular plate so that it can be rotated for making angular cuts. It can swivel from 0° to 90° in a vertical plane. The tool head can be raised or lowered by hand feed for vertical cuts on the work piece by its hand crank for precise depth adjustments.

Clapper Box:

The clapper box is needed because the cutter drags over the work on the return stroke. The clapper box is hinged so that the cutting tool will not dig in. Often this clapper box is automatically raised by mechanical, air, or hydraulic action.

Cross Rail:

The Cross rail is a heavy casting attached to the column at its front on the vertical guide ways. It carries the horizontal table slide ways. The cross rail can be raised or lowered by means of an elevating screw in order to compensate for different thicknesses of work.

Table: It is made of cast iron and has box type construction. It holds and supports the work during the operation and slides along the cross rail to provide feed to the work. T-slots are provided on its top and sides for securing the work to it. The table is moved left and right, usually by hand, to position the work under the cutter when setting up. Then, either by hand or more often automatically, the table is moved sideways to feed the work under the cutter at the end or beginning of each stroke.

Saddle:

The saddle moves up and down (Y axis), usually manually, to set the rough position of the depth of cut. Final depth can be set by the hand crank on the tool head.

Tool holders:

Tool holders are the same as the ones used on an engine lathe, though often larger in size. The cutter is sharpened with rake and clearance angles similar to lathe tools though the angles are smaller because the work surface is usually flat. These cutters are fastened into the tool holder.

Work holding:

Work holding is frequently done in a vise. The vise is specially designed for use in shapers and has long ways which allow the jaws to open up to 14" or more, therefore quite large work pieces can be held. The vise may also have a swivel base so that cuts may be made at an angle. Work that cannot be held in the vise (due to size or shape) is clamped directly to the shaper table in much the same way as parts are secured on milling machine tables.

1.3 Classification of Shapers:

According to the reciprocating motion to the ram:

- Crank type
- Gear type
- Hydraulic type

1.3.1 Quick Return Mechanism

This is the most common type of shaper in which a single point cutting tool is given a reciprocating equal to the length of the stroke desired while the work is clamped in position on an adjustable table. In construction, the crank shaper employs a crank mechanism to change circular motion of a large gear called "bull gear" incorporated in the machine to reciprocating motion of the ram. The bull gear receives power either from an individual motor or from an overhead line shaft if it is a belt driven shaper. The shaping machine is used to machine flat metal surfaces. The reciprocating motion of the mechanism inside the shaping machine can be seen in the diagram. As the disc rotates the top of the machine moves forwards and backwards, pushing a cutting tool. The cutting tool removes the metal from work which is carefully bolted down.

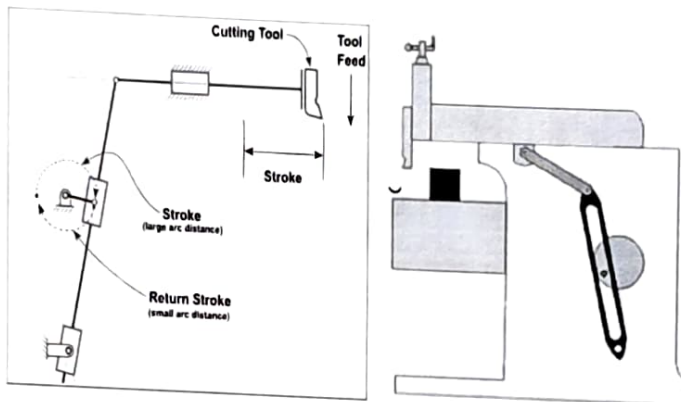


Fig.2 Quick return mechanism for shaper

1.3.2 Geared Type

The reciprocating motion of the ram in some type of shaper is affected by means of a rack and pinion. The rack teeth which are cut directly below the ram mesh with a spur gear. The pinion meshing with the rack is driven by a gear train. The speed and the direction in which the machine will traverse depend on the number of gears in the gear train. This type of shaper is not very widely used.

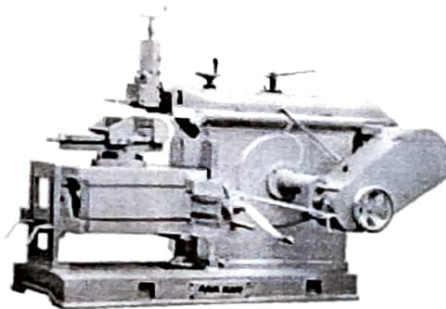


Fig.3 Gear Shaping Machine

1.3.3 Hydraulic Mechanism

In a hydraulic shaper, reciprocating movement of the ram is obtained by hydraulic power. Oil under high pressure is pumped into the operating cylinder fitted with a piston. The end of the piston rod is connected to the ram. The high pressure oil first acts on one side of the piston and then on the other causing the piston to reciprocate and the motion is transmitted to the ram. The piston speed is changed by varying the amount of liquid delivered by the pump. One of the most important advantages of this type of shaper is that the cutting speed and force of the ram drive are constant from the very beginning to the end of the cut. It also offers great flexibility of speed and feed control., eliminates shock and permits slip or slowing up of motion when the cutting tool is overloaded, protecting the parts or the tools from breakage. Another advantage is that the machine does not make any noise and operates very quietly.

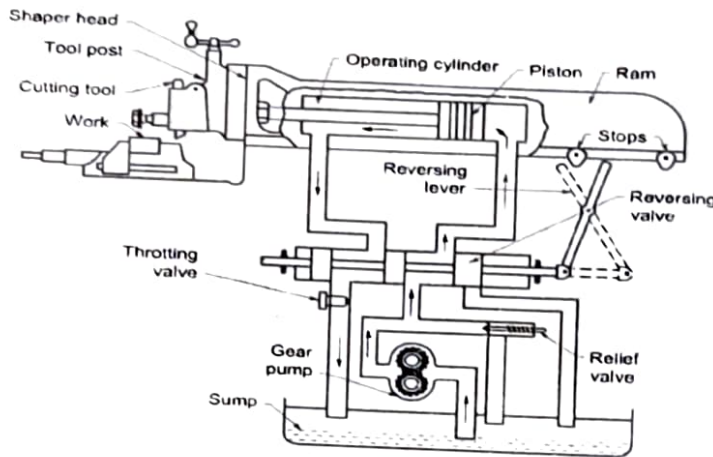


Fig.4 Hydraulic Drive for a shaper

1.4 According to the position and travel of the ram:

Horizontal type

Vertical type

Travelling head type

1.4.1 Horizontal Shaper:

In a horizontal shaper, the ram holding the tool reciprocates in a horizontal axis. Horizontal shapers are mainly used to produce flat surfaces.

1.4.2 Vertical Shaper:

In a vertical shaper, the ram holding the tool reciprocates in a vertical axis. In some of the vertical machines provision is made to allow adjustment of the ram to an angle of about 10 degrees from the vertical position. Vertical shapers may be crank driven, rack driven, screw driven or hydraulic power driven. The work table of a vertical shaper can be given cross, longitudinal, and rotary movement. The tool used on a vertical shaper is entirely different from that used on a horizontal shaper. Vertical shapers are very convenient for machining internal surface, keyways, slots or grooves. Large internal and external gears may also be machined by indexing arrangement of the rotary table. There

are vertical shapers which are specially designed for machining internal keyways. They are then called key seaters.

1.4.3 Travelling Shaper:

In a traveling head shaper, the ram carrying the tool while it reciprocates moves crosswise to give the required feed. Heavy and unwieldy jobs which are very difficult to hold on the table of a standard fed past the tool are held static on the basement of the machine while the reciprocates and supplies the feeding movements.

1.5 According to the type of design to the table:

Standard shaper

Universal shaper

1.5.1 Standard shaper

A shaper is termed as standard or plain when the table has only two movements, vertical and horizontal, to give the feed. The table may or may not be supported at the outer end.

1.5.2 Universal Shaper:

In a universal shaper, in addition to the two movements provided on the table of a standard shaper, the table can be swiveled about an axis parallel to the ram ways, and upper portion of the table can be tilted about a second horizontal axis perpendicular to the first axis. As the work mounted on the table can be adjusted in different planes, the machine is most suitable for different types of work and is given name "Universal". A universal is mostly used in tool room work.

1.6 According to the type of cutting stroke:

Push type

Draw type

1.6.1 Push Type Shaper:

This is the most general type of shaper used in common practice. The metal is removed when the ram moves away from the column, i.e. pushes the work.

1.6.2 Draw Type Shaper:

In a draw shaper, the metal is removed when the ram moves towards the column of the machine, i.e., draws the work towards the machine. The tool is set in a reversed direction to that of a standard shaper. The ram is generally supported by an overhead arm which ensures rigidity and eliminates deflection of the tool. In this shaper the cutting pressure acts towards the column which relieves the cross rail and other bearings from excessive loading and allows taking deep cuts. Vibration in these machines is practically eliminated.

1.7 Shaper Specifications:

Shaper machine are manufactured by taking account of the following specifications:

1. Length of Stroke
2. Maximum Horizontal Travel of Table
3. Maximum Vertical Travel of Table
4. Maximum distance from Table to Ram
5. Maximum Vertical Travel of Tool Slide
6. Length & Width of Table Top

7. Length & Depth of Table Side

8. Power of Motor

1.8 Operations on Shaper:

Machining Horizontal Surface:

Set the vice jaws length perpendicular to the ram movement. Hold the job lengthwise parallel to the tool stroke so that maximum stroke of the tool may be utilized. Hold the appropriate tool in tool head. Set the proper inclination of the tool and depth of cut. Give cross feed to the tool initially by hand till the cut start and then employ power feed.

Repeat the above said procedure.

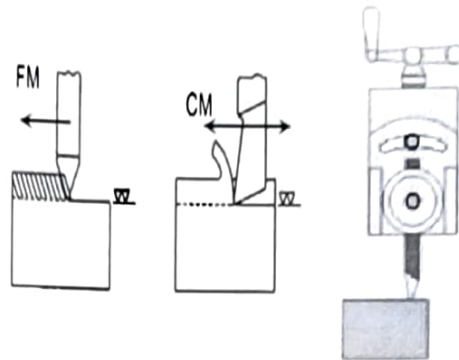


Fig.5 Horizontal Shaping

Machining Vertical Surface:

The tool is fed downward in vertical cutting so swivel the clapper box from the face of the work to be cut. The tool is fed downwards by rotating the down feed screw by hand at the end of return stroke. On the return stroke the tool swings away from the work and gives clearance, which prevents the work from being scratched. Start cutting from highest point towards the bottom.

Repeat the procedure for another cut.

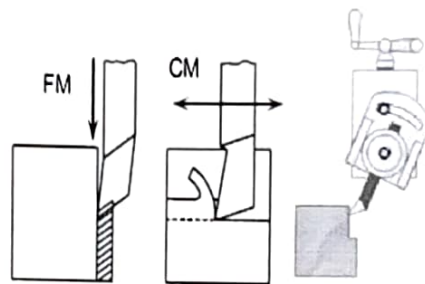


Fig.6 Vertical Shaping

Machining Angular Surfaces:

Set the swivel head to the required angle for shaping angular faces. The clapper box is to be swung away from the face to be machined. The tool is to be fed by the slide hand wheel. The apron top is also swiveled in the direction away from the surface to be machined. Feed the tool during return stroke. Repeat the procedure to finish the work.

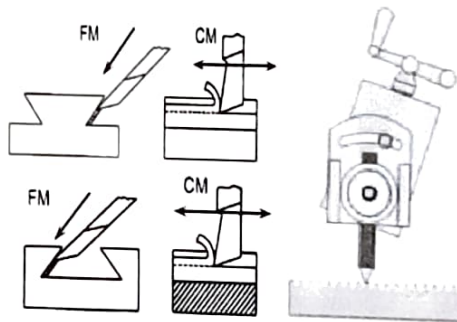


Fig.7 Angular Shaping

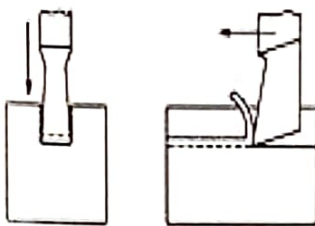


Fig.8 Slotting operation

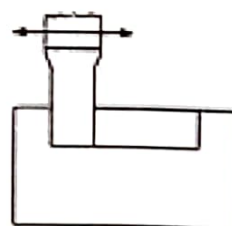


Fig.9 Pocketing operation

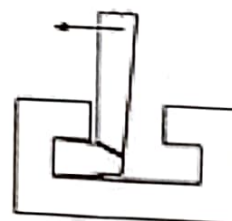


Fig.10 T-Slot cutting

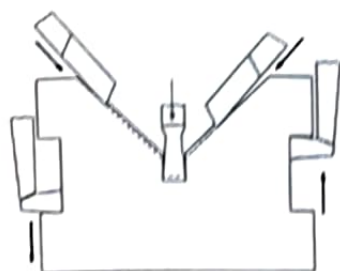


Fig.11 Vee-block Cutting Fig.41 Groove Cutting

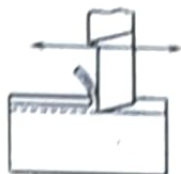


Fig. 12 Spur Gear

- 4 Some unusual work such as slotting operation, pocketing, T-Slot Cutting, Vee - Block Cutting, Grooving, Spur Gear cutting using straight tooth, etc, can also be done, if needed, by developing and using special attachments.

DO'S & DON'TS FOR THE SHAPER

DO'S

- ✓ Do thoroughly understand the requirements of the job.
- ✓ Do see that there is at least 2" clearance between the ram and the vise or work.
- ✓ Do keep your hands away from the moving work & tool holder.
- ✓ Do make sure that the job is held securely.
- ✓ Do keep your mind on the work.
- ✓ Do make sure that the tool head is not down too far.
- ✓ Do see that the shaper is well oiled.

DON'TS

- ✗ Don't walk away from the shaper and leave machine running.
- ✗ Don't stand directly in front of shaper while it is in operation.
- ✗ Don't try to feed the tool head down without loosening the gib friction bolt.
- ✗ Don't forget to increase the speed of the ram before taking a finishing cut. It gives a better finish and completes the job in a shorter time.
- ✗ Don't hammer on the tool head or clapper box to swivel them. Loosen the nuts sufficiently so that the head or box can be turned easily by the hand.